

Nikhil Chaudhary

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PROFESSIONAL SUMMARY

Research Software Engineer with 2+ years of experience building scientific applications, reproducible AI/ML workflows, backend systems, and data-driven software. Proficient in Python, FastAPI, C#/.NET, PostgreSQL, PyTorch, Docker, and CI/CD, with experience in numerical validation, API development, benchmarking, data provenance, and production deployment.

SKILLS

Programming: Python, C#, TypeScript, JavaScript, SQL, Java, C++
Research Software: Scientific Computing, Reproducible Workflows, Benchmarking, Automated Testing, API Design
AI/ML & Data: Machine Learning, RAG, Model Evaluation, PyTorch, TensorFlow, scikit-learn, Pandas, NumPy
Frameworks & Tools: FastAPI, .NET, Angular, REST APIs, PostgreSQL, MySQL, SQLite, ChromaDB, Docker, Git, CI/CD, Linux

EXPERIENCE

Software Developer

Sep 2024 – May 2026

South Dakota State University — Structural Engineering Research

Brookings, SD

- Built a C#/.NET structural-analysis application backed by SQLite for Maguire Iron, automating previously manual engineering load calculations and cutting per-project calculation time by an estimated 60%.
- Eliminated manual computation errors by validating outputs against verified benchmarks, and deployed a self-contained Windows installer for non-technical engineers.

Software Development Engineer

Jul 2022 – Jul 2024

Siemens Technologies

Gurgaon, India

- Developed a full-stack municipal water-supply management platform using Angular, .NET, REST APIs, PostgreSQL, and Docker, deployed to production for municipal operations.
- Migrated legacy AngularJS modules to Angular 14, reducing front-end load times by nearly 30% and improving long-term maintainability.
- Designed REST APIs and Dockerized services in an Agile team, improving deployment reliability and cutting environment-setup time on a 2-week release cadence.

Data Science Intern

Dec 2020 – Jun 2021

HighRadius Technologies

Bhubaneswar, India

- Performed exploratory data analysis, feature engineering, and ML model evaluation on financial datasets using Python, Pandas, and scikit-learn to support predictive-modeling initiatives.

PROJECTS

PolicyProof — Reproducible RAG Evaluation System

In Progress

github.com/Bad33/policyproof

Python, PyTorch, ONNX, BM25, Dense Retrieval, GitHub Actions

- Building a reproducible RAG and citation-verification pipeline for public AI-governance documents with source provenance, deterministic retrieval evaluation, abstention handling, and regression testing.
- Implemented corpus ingestion, token-safe passage construction, benchmark datasets, BM25 and dense-retrieval baselines, evaluation metrics, and hash-verified generated artifacts.
- Documented evaluation methodology, engineering decisions, model assets, and failure analysis to support maintainability and reproducible experimentation.

ClaimPilot — Explainable AI Claims Triage Platform

github.com/Bad33/claimpilot

Python, FastAPI, OpenAI API, scikit-learn, PostgreSQL, Docker

- Developed an AI document-intelligence platform for claims ingestion, extraction, and triage, returning evidence snippets, confidence scores, reason codes, and audit logs for traceable decisions.
- Built FastAPI services and PostgreSQL-backed workflows for document processing and audit-log management, and containerized the application with Docker for reproducible deployment.

RESEARCH PUBLICATIONS

[GOOGLE SCHOLAR](#)

Co-authored peer-reviewed research involving machine-learning mortality prediction, FAERS pharmacovigilance, and geographic access to cancer clinical trials, published or accepted in *Blood*, *Journal of Clinical Oncology*, and *Clinical Lymphoma, Myeloma and Leukemia*.

EDUCATION

M.S. in Computer Science, South Dakota State University

Brookings, SD | Aug 2024 – May 2026

B.Tech in Computer Science and Engineering, KIIT University

Bhubaneswar, India | Jul 2017 – Aug 2021